Decision Making Statements

Monday, January 8, 2024 8:51 AM

As the name suggests, decision-making statements decide which statement to execute and when. Decision-making statements evaluate the Boolean expression and control the program flow depending upon the result of the condition provided. There are two types of decision-making statements in Java i.e., :

1. if statement
2. switch statement

1. **if statement:** In Java, the "if" statement is used to evaluate a condition. The control of the program is diverted upon the specific condition. The condition of the if statement gives a Boolean value, either true or false.

   Syntax:

```
if (<condition>){
        //code to execute when the condition is true;
}
```

   For example:

```
import java.util.*;
public class DemoOfIf
{
   public static void main(String args[]){
      Scanner sc = new Scanner(System.in);
      int age;
      System.out.print("\fEnter your age: ");
      age = sc.nextInt();
      if(age >= 18){
         System.out.print("\nYou are elligible to learn driving." );
      }
   }
}
```

if-else statement:

The if-else statement is an extension to the if-statement, which uses another block of code, i.e., else block. The else block is executed if the condition of the if-statement is false.

Syntax:

```
    if(<condition>){
        //code to execute when above condition is true.
    }
    else{
        //code to execute when above condition is false.
    }
```

For example:

```
import java.util.*;

public class DemoOfIf
{
   public static void main(String args[]){
      Scanner sc = new Scanner(System.in);
      int age;
      System.out.print("\fEnter your age: ");
      age = sc.nextInt();
      if(age >= 18){
         System.out.print("\nYou are elligible to learn driving." );
      }
      else{
         System.out.print("\nYou are not elligible to learn driving.");
      }
   }
}
```

if-else-if ladder: the if-else-if statement contains the if-statement followed by multiple else-if statement. In other words, we can say that it is the chain of if-else statements that create a decision tree where the program may enter in the block of code where the condition is true. We can also define else statement at the end of the chain.

Syntax:

```
if(<condition>){
    //code to be executed if the above condition is true.
}
else if(<condition>){
    //code to be executed if the else-if condition is true.
}
else if(condition>){
    //code to be executed if the else-if condition is true.
}
else{
    //code to be executed if the else-if condition is false.
}
```

9. A Pre-Paid taxi charges from the passenger as per the tariff given below:

Distance	Rate
Up to 5 km	₹ 100
For the next 10 km	₹ 10/km
For the next 10 km	₹ 8/km
More than 25 km	₹ 5/km

Write a program to input the distance covered and calculate the amount to be paid by the passenger. The program displays the printed bill with the details given below:

Taxi No. : _____

Distance covered : _____

Amount : _____

festival discounts and the gifts on the

```
package Chapter3;
import java.util.*;


/**
* Write a description of class Q9 here.
*
* @author (your name)
* @version (a version number or a date)
*/
```

```
public class Q9
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        int dis, amt;
        String taxiNum;
        System.out.print("\fEnter taxi number: ");
        taxiNum = sc.next();
        System.out.print("Enter distance: ");
        dis = sc.nextInt();
        if(dis <= 5){
            amt = 100;
        }
        else if (dis <= 15){
            amt = 100 + (dis - 5) * 10;
        }
        else if (dis <= 25){
            amt = 200 + (dis - 15) * 8;
        }
        else{
            amt = 280 + (dis - 25) * 5;
        }
        System.out.print("\n************BILL************");
        System.out.print("\nTaxi No. " + taxiNum);
        System.out.print("\nDistance covered: " + dis);
        System.out.print("\nAmount to paid: " + amt);
        System.out.print("\n**********Thank you************");
    }
}
```

Nested if-statement: the nested if-statements can contain if or if-else statement inside another if or if-else statement.

Syntax:

```
 if (<condition>){
     //
     if(<condition>){
         //
```

```
    if(<condition>){
        //
    }
    else{
        //
    }
}
else{
    //
    if(<condition>){
        //
    }
    else{
        //
    }
}
```

← nested or inside.

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For example:

1. Write a program to input three angles of a triangle and check whether its construction is possible or not. If possible then check and display whether it is an acute-angled triangle, right-angled or an obtuse-angled triangle. Otherwise, display 'A Triangle is not possible'.
   Sample Input: Enter three angles: 40, 50, 90
   Sample Output: Right–Angled Triangle

```
import java.util.*;


public class Q1
{
   public static void main(String args[]){
      Scanner sc = new Scanner(System.in);
      int a, b, c, s;
      System.out.print("\fEnter first angle: ");
      a = sc.nextInt();
      System.out.print("Enter second angle: ");
```

```
        b = sc.nextInt();
        System.out.print("Enter third angle: ");
        c = sc.nextInt();
        if(a <= 0 || b <= 0 || c <= 0){
            System.out.print("\nTriangle is not possible!");
        }
        else{
            s = a + b + c;
            if(s == 180){
                if(a == 90 || b == 90 || c == 90){
                    System.out.print("\nRight angled triangle.");
                }
                else if(a < 90 && b < 90 && c < 90){
                    System.out.print("\nAcute angled triangle.");
                }
                else{
                    System.out.print("\nObtused angled triangle.");
                }
            }
            else{
                System.out.print("\nConstruction of triangle is not possible.");
            }
        }
    }
}
```